F24-011-D-SmartGuard

Project Team

Umer Farooq	21I-1184
Harris Maizan	21I-1146
Abdullah Khan	21I-1176

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Supervised by

Dr. Asif Muhammad

Co-Supervised by

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Department of Computer Science

National University of Computer and Emerging Sciences
Islamabad, Pakistan

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Chapter 1

Introduction

The following document specifies the requirements of Smart Guard,an AI-based shoplifting detection system. This document is intended for various readers including Smart Guard development team, Smart Guard FYP supervisor and panel members. [4].

1.1 Problem Statement

According to a survey conducted by the National Retail Federation (NFR), retail theft recorded in 2017 was over 112 billion dollars, and this amount is projected to rise above 140 billion dollars by 2025 [4]. Retail theft is a looming vice that leads to major losses and challenges to retail security. Traditional security measures such as CCTV surveillance systems are, most of the time, irrelevant because people are prone to failures. Such shortcomings make it hard for the retailer to respond and do things at the right time. Therefore, there is a growing need to develop an automated system for detecting shoplifting activities and alerting store management.

1.2 Scope

In-Scope

The scope of the **Smart Guard** project is to identify certain shoplifting behaviors using computer vision. The system is intended to recognize actions that include the following:

- Picking a product from the shelf.
- Examining a product.

- Putting the product back on the shelf after reviewing it.
- Placing a product in a bag, pocket, or clothing without paying for it.

These behaviors are tracked in real time, and whenever any suspicious activity, such as shoplifting, is noticed, an alert is sent to the store staff to handle the incident.

Out-of-Scope

The project does not cover other criminal behaviors, such as:

- Physical aggression.
- Property damage.
- Verbal aggression.

Additionally, it does not address cases where people consume products in the store (e.g., eating or drinking from shelves without paying). The system's effectiveness may be limited in situations where faces are obscured by masks, clothing, or any other obstructions, as facial recognition requires clear visibility of a person's face

1.3 Modules

1.3.1 Real-Time Monitoring and Detection

This module continuously monitors live camera feeds to detect suspicious activities, and in case of any suspicious activity it sends an alert to store management so that they can take any action.

1. Live Feed Monitoring.
2. Suspicious Behavior Detection.
3. Alert Management.

1.3.2 Facial Recognition and Centralized Watch list

Utilizes facial recognition to identify individuals from a centralized watch list, helping store management track known offenders across multiple stores. The system will continuously train on the pictures stored in the centralized watch list.

1. Facial Recognition.

2. Common Watch list.

1.3.3 Customer Insights and Analytics

To enhance store security and inform management decisions, track customer visit frequency and log shoplifting incidents, providing data driven reports. .

1. Visit Count

2. Incident Reporting and Analytics.

1.3.4 Advanced Search and Store Management

Enables advanced search capabilities using image and filters, while also managing all the stores connected to Smart Guard.

1. Search by image

2. Advance Search.

3. Store Management.

1.4 User Classes and Characteristics

User class	Description
Store Managers	Receive real-time alerts,access reports and analytics via mobile or web application.
Store Employees	Monitor camera feed, respond to real-time alerts of shoplifting,provide on-the-ground support during any incident.
System Administrators	Handle system,Add and remove stores, provide technical support.
CCTV Technicians	Ensure proper integration of CCTV cameras with Smart Guard, and handle hardware maintenance.

Chapter 2

Project Requirements

2.1 Event Response Table

Event	System State	System Response
New video frame received from camera feed	Processing frame, detecting objects and activities	Process the frame using AI algorithms. Detect objects and human activities.
Suspicious behavior pattern detected	Event logged, notifications triggered	Log the event with timestamp and details. Send notifications to mobile and web applications.
Known offender recognized via facial recognition	Cross-referencing watch list database, sending alert if match is found	Cross-reference face with watch list database. If a match is found, send alert to store management.
New individual added to the watch list	Centralized database updated, facial recognition model retrained	Update the centralized database. Retrain facial recognition model with new data.
User searches for an individual using an image	Image processed, comparison performed, results returned or notification sent	Process the uploaded image. Compare against images stored in common watchlist. Return matching profiles or notify if none is found.
User logs into the system	Authenticating credentials, access granted to respective stores	Authenticate credentials and grant access to their respective stores.
Incident report requested	Data retrieved, report generated, display options provided	Retrieve relevant data from the database. Generate report in the requested format. Display report to the user. Offer options to export or print.
New store added to the system	Store database updated, resources allocated	Update the store's database. Allocate resources for the new store.

Table 2.1: Event Response Table

2.2 Functional Requirements

2.2.1 Real Time Monitoring and Detection

This module continuously monitors live camera feeds to detect suspicious activities, and in case of any suspicious activity it sends an alert to store management so that they can take any action.

1. FR1 :The system must continuously monitor the connected cameras to detect suspicious behavior.
2. FR2 :The system should detect suspicious behavior like picking an item from the shelf and putting it in a bag.
3. FR3 :The system should send a notification to the store management if suspicious behavior is detected.

2.2.2 Facial recognition and centralized watchlist

Utilizes facial recognition to identify individuals from a centralized watch list, helping store management track known offenders across multiple stores. The system will continuously train on the pictures stored in the centralized watch list

1. FR1: The system should perform facial recognition on the person detected by the camera.
2. FR2 :The system should compare the faces detected by the camera with the common watchlist to detect known offenders.
3. FR3 :The watchlist should be accessible by all the stores.
4. FR4 :The system should send a notification once a known shoplifter is detected.

2.2.3 Customer Insights and Analytics

To enhance store security and inform management decisions, track customer visit frequency and log shoplifting incidents, providing data driven reports.

1. FR1 : The system records the number of customers visiting the store.
2. FR2 : The system should log shoplifting incidents, including the date, time and store location.

2.2.4 Advance search and store management

Enables advanced search capabilities using image and filters, while also managing all the stores connected to Smart Guard.

1. FR1 : The system should allow store management to search a person by image based input.
2. FR2 : The system should allow store managers to manage multiple stores from a single interface.

2.3 Non-Functional Requirements

2.3.1 Reliability

1. REL-1: The system should achieve an MTBF (Mean Time Between Failures) of at least 1,000 hours of operation
2. REL-2: The system should ensure data integrity during failures, with no loss of recorded data, logs, or alerts.

2.3.2 Usability

1. USE-1: The system should allow a user to access core functions (e.g., viewing camera feeds, responding to alerts) with no more than 3 clicks.
2. USE-2: Users must be able to operate the system after no more than 2 hours of training.
3. USE-3: The mobile application should be developed exclusively for Android devices, ensuring compatibility with Android versions 10 (Android Q) [2] and above
4. USE-4: The web interface should work across all modern browsers, including Chrome, Firefox, Safari, and Edge.

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2.3.3 Performance

1. PER-1: Live video feeds should have a latency of less than 8 second, ensuring near real-time monitoring.

2. PER-2: Detection of suspicious activity must occur within 10 seconds of the event being captured on camera.
3. PER-3: The system should be able to handle up to 50 concurrent users accessing the web interface without noticeable lag or delays.
4. PER-4: Facial recognition searches should return results in less than 5 seconds when scanning against the centralized watch list.
5. PER-5: The system should process and send alerts within 10 seconds of detecting suspicious behavior.

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2.3.4 Security

1. SEC-1: Data, including shop details, passwords, should be encrypted at rest and in transit using at least 256-bit AES encryption [1].
2. SEC-2: After 3 failed login attempts, the account should be locked for 10 minutes to prevent brute-force attacks.
3. SEC-3: The system should log all user actions, including login attempts, access to critical features, and with logs stored for 1 year.

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2.3.5 Scalability

1. SCA-1: The system should be scalable to handle up to 2 stores connected to a centralized watch list, with each store running up to 3 cameras.
2. SCA-2: The cloud infrastructure should automatically scale based on system load, scaling up when CPU utilization exceeds 70% and scaling down when utilization drops below 30%, using discrete scaling thresholds

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2.4 Domain Model

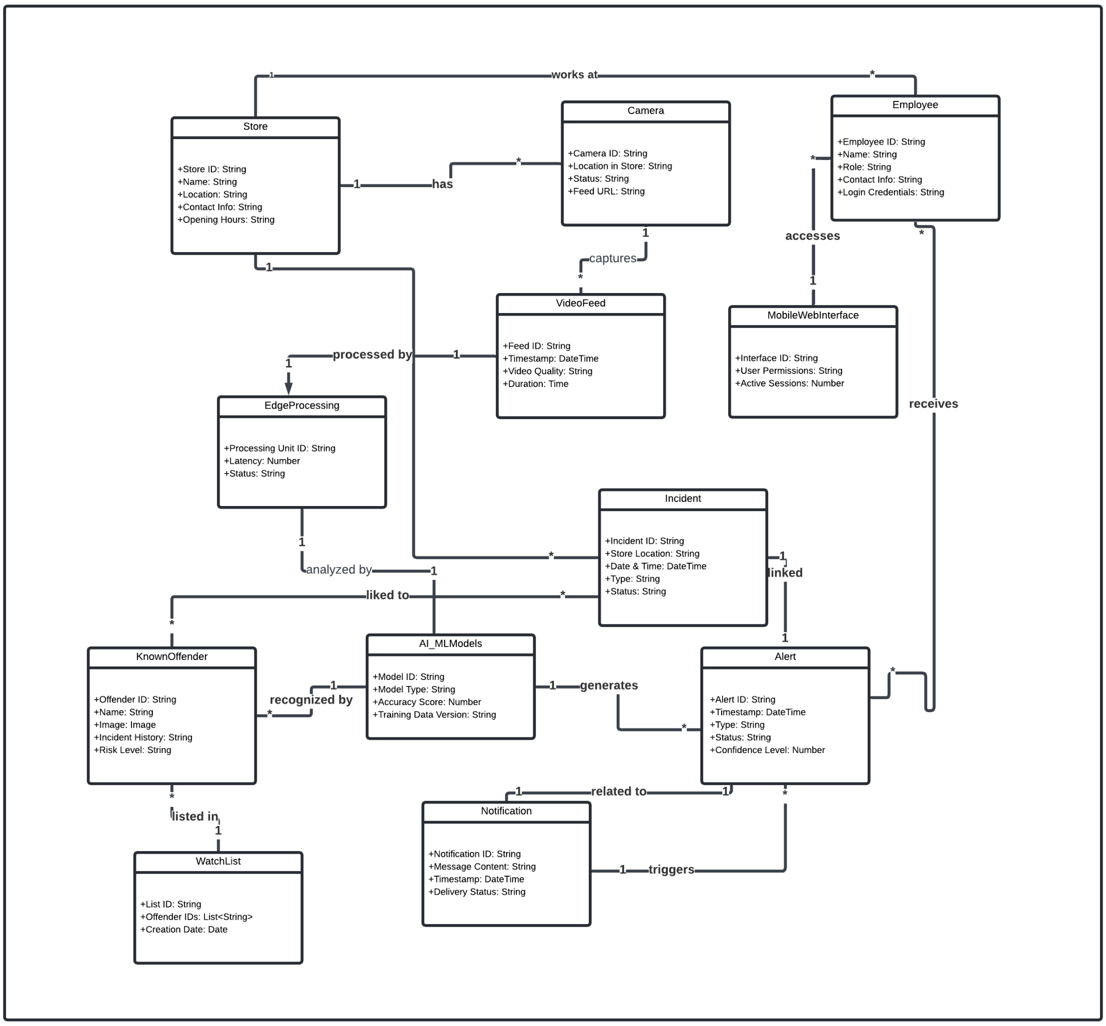

Figure 2.1: Domain Model Representation

2.5 Architectural Design

The architectural design provides a modular program structure to achieve the complete functionality of the system. The aim is to give a high-level overview of how the system is decomposed into modules and how these modules work together.

2.5.1 Initial Modular Structure Overview

The Smart Guard system consists of the following core modules:

1. **Edge Devices Module**
   - Captures real-time video feeds and pre-processes the data.
2. **Edge Processing Module**
   - Performs initial AI-based analysis on the edge devices for a quick response.
3. **Backend Services Module**
   - Contains services for deeper AI analysis, data management, alerting, and incident tracking.
4. **AI/ML Models Module**
   - Provides behavior detection, facial recognition, and data analytics capabilities.
5. **Database Module**
   - Stores data like user information, watch lists, incidents, and analytics.
6. **Notification & Alerting Module**
   - Manages the dispatch of real-time alerts and notifications to web and mobile applications.
7. **Web and Mobile Interface Module**
   - Allows store managers and security personnel to interact with the system, view live feeds, and receive alerts.

2.5.2 Box and Line Diagram

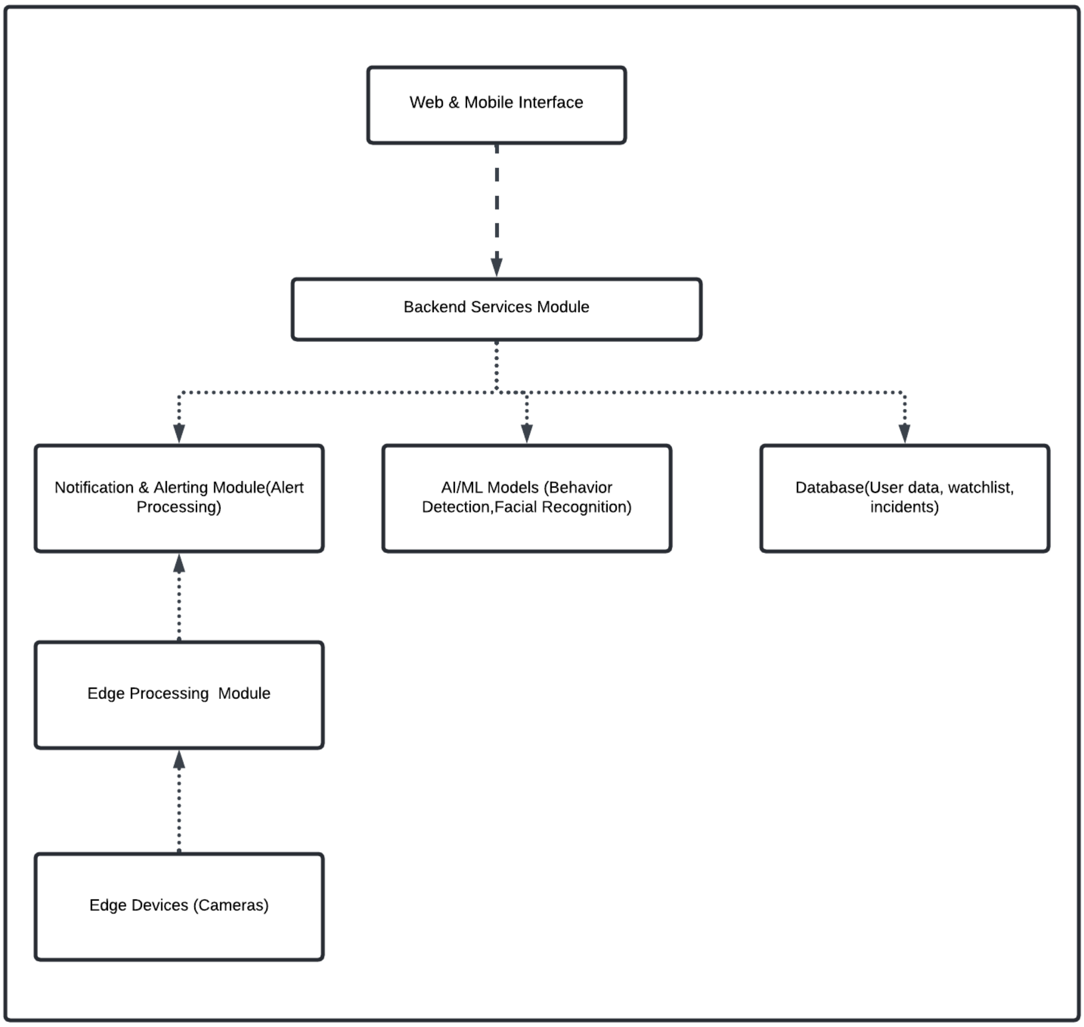

Figure 2.2: Box and Line Diagram of Smart Guard System

2.5.3 Multi-Tiered (Layered) Architecture

The Smart Guard system uses a multi-tiered layered [3] architecture. This architecture separates modules into distinct layers to ensure a clean separation of concerns. The benefits include scalability, maintainability, and efficient real-time processing.

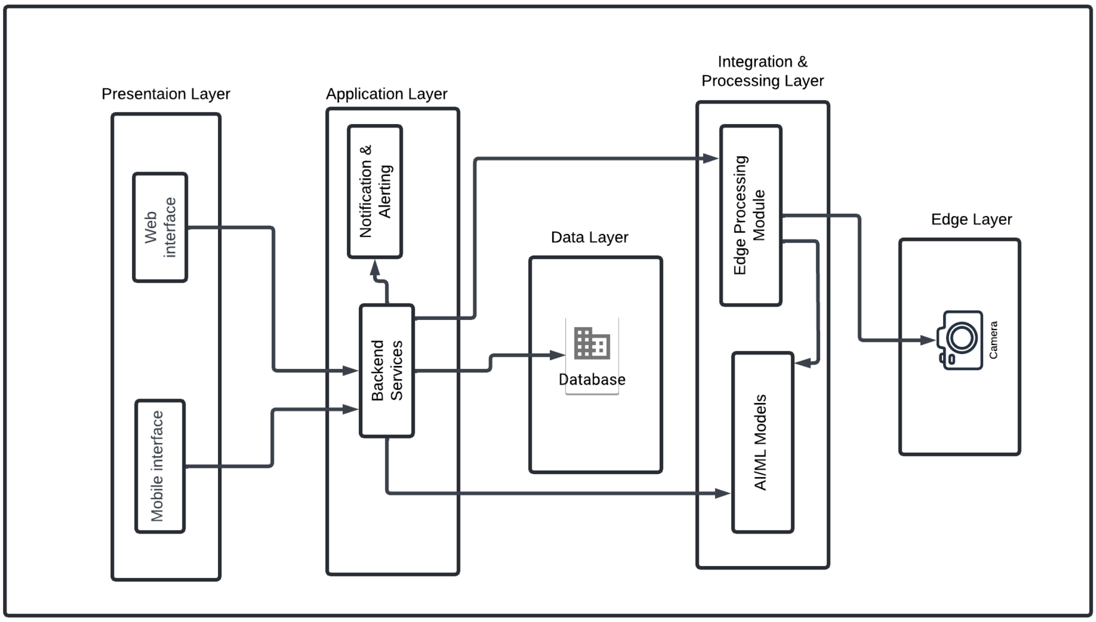

Figure 2.3: Architecture Diagram

2.5.4 Detailed Mapping of Modules to Architecture Layers

2.5.4.1 Presentation Layer (UI Layer)

- **Modules**: Web & Mobile Interface
- **Responsibilities**: Provides a user-friendly interface for accessing system features, viewing live feeds, and managing incidents and alerts.

2.5.4.2 Application Layer (Business Logic Layer)

- **Modules**: Backend Services, Notification & Alerting
- **Responsibilities**: Handles core functionality, processes requests from the UI, coordinates between other layers, and manages notifications.

2.5.4.3 Data Layer (Data Access Layer)

- **Modules**: Database
- **Responsibilities**: Stores all data related to users, incidents, watch lists, and system configurations. Ensures secure storage and efficient retrieval of data.

2.5.4.4 Integration & Processing Layer

- **Modules**: Edge Processing, AI/ML Models
- **Responsibilities**:
  - **Edge Processing**: Handles initial data analysis to minimize latency.
  - **AI/ML Models**: Performs complex processing tasks such as behavior detection, facial recognition, model training, and analytics.

2.5.4.5 Edge Layer (Device Layer)

- **Modules**: Edge Devices
- **Responsibilities**: Captures video feeds and transmits data for further processing and analysis.

2.6 Design Models

2.6.1 Activity Diagram

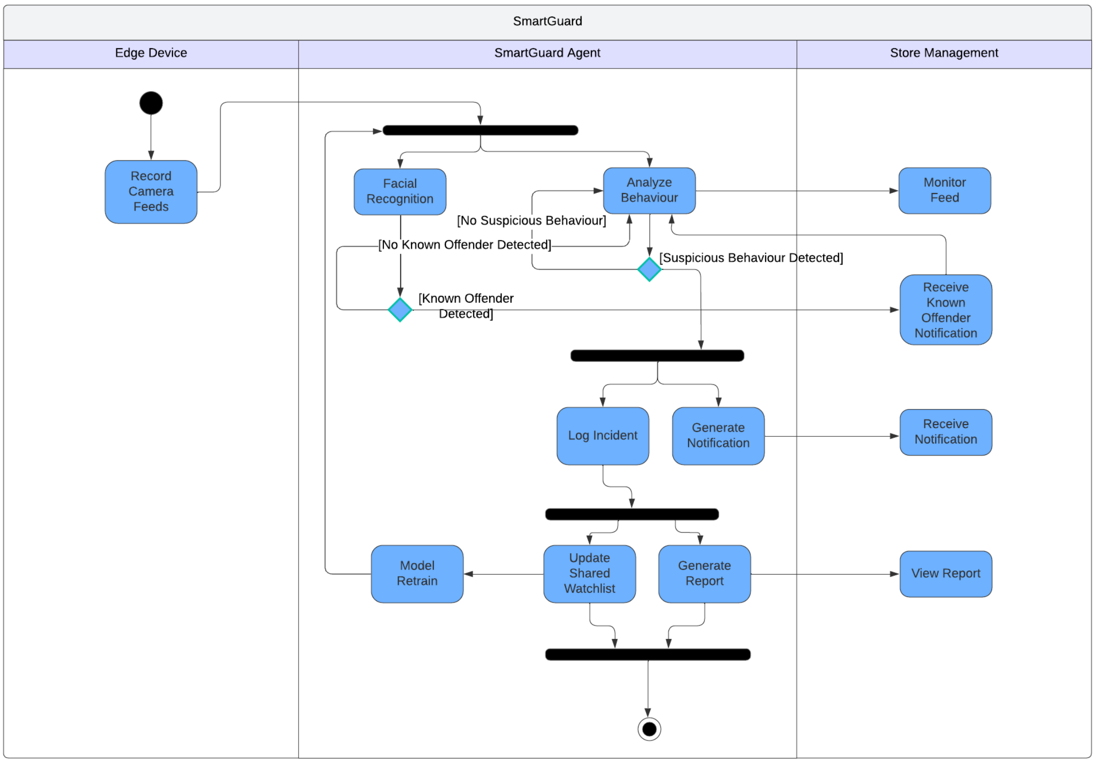

Figure 2.4: Activity Diagram

2.6.2 Data Flow Diagram Level 0

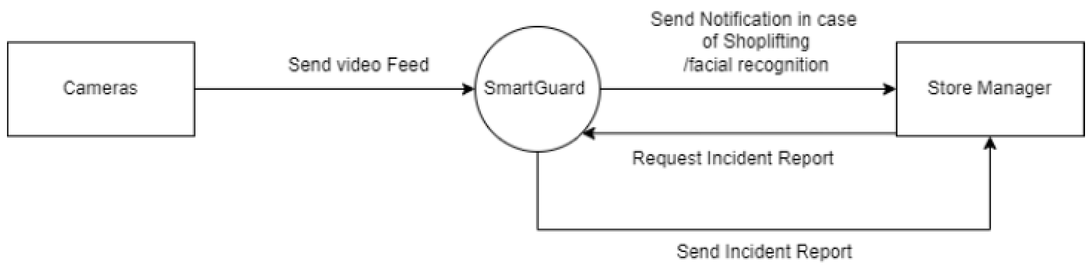

Figure 2.5: Data Flow Diagram Level 0

2.6.3 Data Flow Diagram Level 1

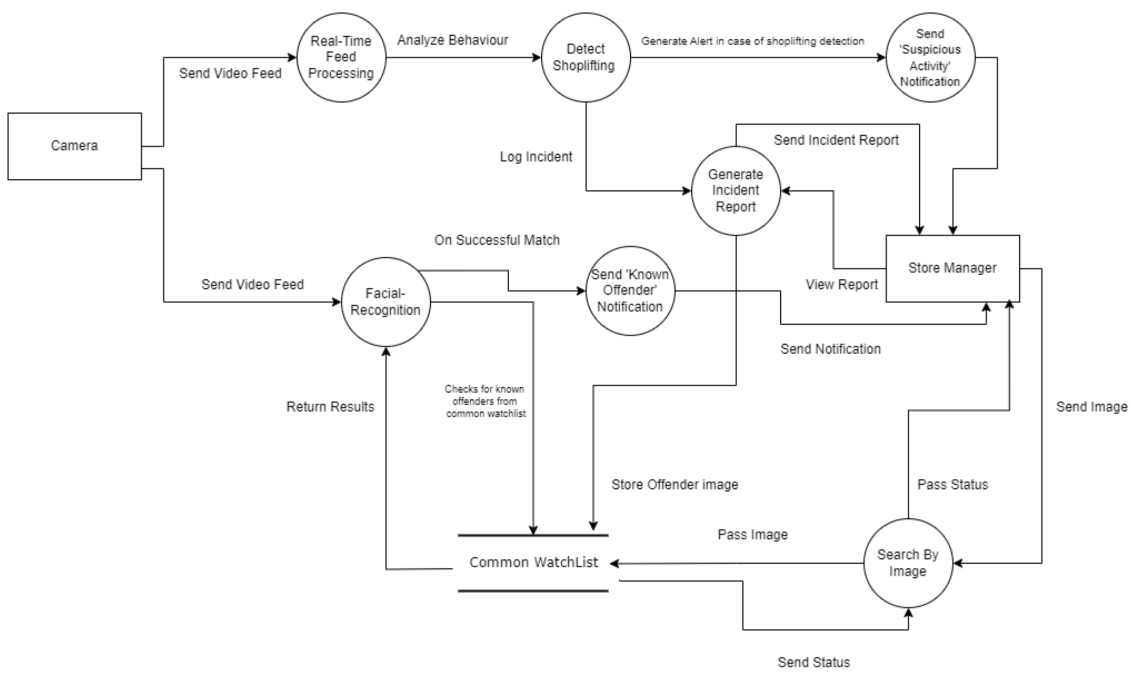

Figure 2.6: Data Flow Diagram Level 1

2.6.4 Data Flow Diagram Level 2

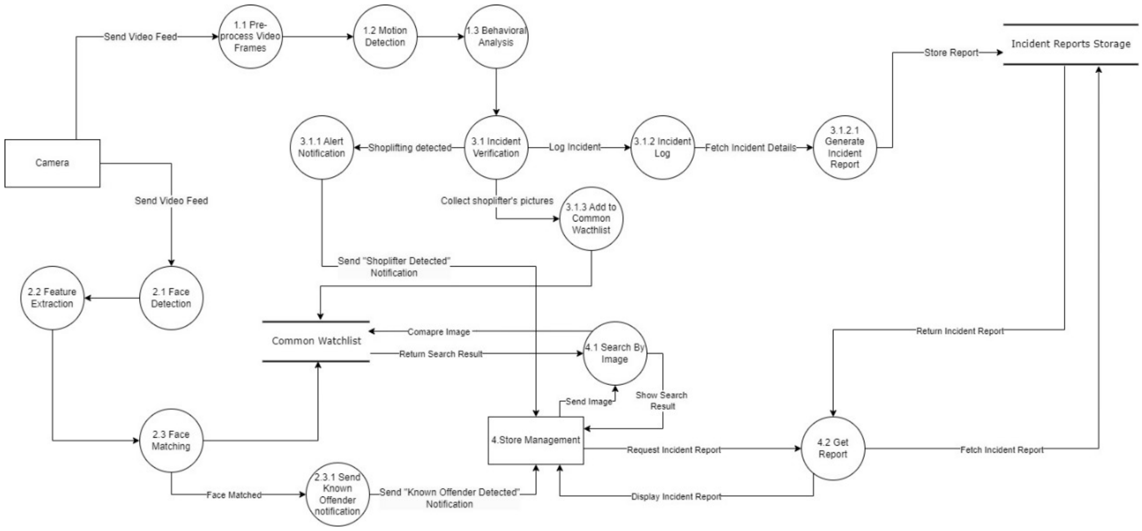

Figure 2.7: Data Flow Diagram Level 2

2.6.5 System Level Sequence Diagram

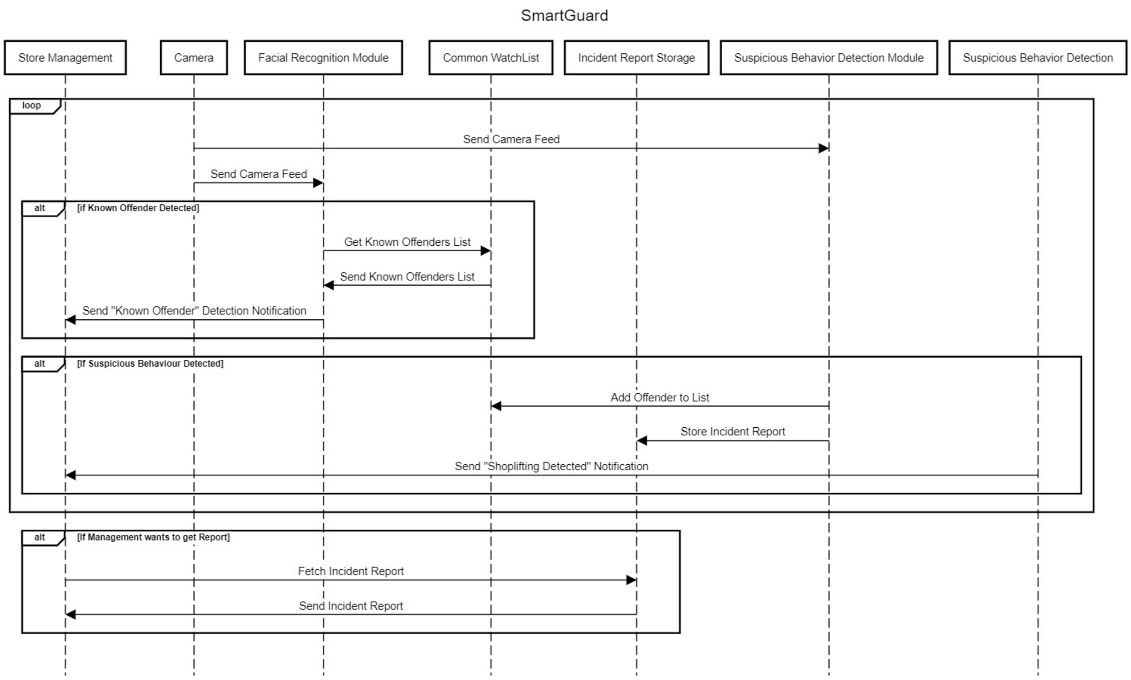

Figure 2.8: System Level Sequence Diagram

2.7 Data Design

The SmartGuard system manages various types of data, including real-time video feed analysis, facial recognition, incident tracking, and store management. The data is organized in a way to ensure fast access, efficient processing, and centralized management. Given the NoSQL storage format, data is stored in JSON documents across collections. Below is a breakdown of how the main data entities are structured and organized:

2.7.1 Store Collection

This collection holds the information about each store connected to the SmartGuard system.

```json
{
  "storeId": "store_001",
  "storeName": "Main Retail Branch",
  "location": {
    "address": "123 Main Street, City, Country",
    "geoCoordinates": {
      "lat": 40.7128,
      "long": -74.0060
    }
  },
  "connectedCameras": [
    "cam_001",
    "cam_002",
    "cam_003"
  ],
  "managers": [
    {
      "managerId": "manager_001",
      "name": "John Doe",
      "contact": "+1234567890",
      "email": "johndoe@example.com"
    }
  ],
  "watchlist": [
    "person_001",
    "person_002"
  ],

"createdAt": "2024-09-01T10:00:00Z",
"updatedAt": "2024-09-30T12:00:00Z"


}
```


Table 2.2: JSON schema of Store Collection

2.7.2 Watchlist Collection

This collection keeps track of known offenders and persons of interest for all stores.


```
{
  "personId": "person_001",
  "fullName": "Jane Smith",
  "photoUrl": "https://example.com/photos/jane_smith.jpg",
  "associatedStores": [
    "store_001",
    "store_002"
  ],
  "addedBy": "manager_001",
  "addedAt": "2024-09-15T09:00:00Z",
  "description": "Known for repeat shoplifting in electronics section",
   "status": "active"
   }
```


Table 2.3: JSON schema of Watchlist Collection

2.7.3 Incident Log Collection

Logs any suspicious behavior or incidents detected in real-time.


```json
{
  "incidentId": "incident_001",
  "storeId": "store_001",
  "cameraId": "cam_001",
  "detectedPersonId": "person_001",
  "incidentType": "suspicious_behavior",
  "timestamp": "2024-09-30T13:20:00Z",
  "details": {
    "description": "Person was seen placing an item in bag.",
    "videoClipUrl": "https://example.com/videos/incident_001.mp4"
  },
  "reportedBySystem": true,
  "resolved": false,
  "createdAt": "2024-09-30T13:21:00Z",
  "updatedAt": "2024-09-30T14:00:00Z"
}
```


Table 2.4: JSON schema of Incident Log Collection

2.7.4 Facial Recognition Analytics Collection

Stores details on facial recognition results for tracking individuals against the watchlist.


```
{
  "recognitionId": "recognition_001",
  "storeId": "store_001",
  "cameraId": "cam_001",
  "detectedPerson": {
    "personId": "person_002",
    "fullName": "John Doe",
    "confidenceScore": 0.92
  },
  "timestamp": "2024-09-30T12:50:00Z",
  "recognizedBy": "AI_Model_v1.2",
  "alertSent": true,
  "createdAt": "2024-09-30T12:51:00Z"
}
```


Table 2.5: JSON schema of Facial Recognition Analytics

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